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Genome annotation, comparative genomics and transcriptomic analysis of *Eucalyptus cloeziana* reveal insights into genome evolution and wood formation in *Eucalyptus*

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A B S T R A C T

Eucalyptus, a fast-growing tree species, has significant economic and ecological value. *Eucalyptus cloeziana*, known for its excellent wood properties, is extensively used to produce high-end furniture, highlighting its economic importance. However, the existing genome annotations and gene expression data for *E. cloeziana* are still insufficient for its molecular genetic improvement. Based on the *E. cloeziana* genome assembly data from the NCBI-GenBank, this study systematically annotated its whole genome. Our genome annotation improved the BUSCO completeness score by nearly 10 %, identifying 41,610 genes and 2089 non-coding RNAs. Through comparative transcriptomic analysis across tissues, we characterized the tissue-specific gene expression patterns in *E. cloeziana*, and found that xylem-specific genes were highly enriched in the lignin biosynthesis-related pathway. Molecular evolutionary analyses revealed that *E. cloeziana* and *E. grandis* underwent whole-genome duplication, indicating the evolution of *Eucalyptus* species. Gene family analysis revealed significant expansion in *E. cloeziana* gene families involved in secondary cell wall and lignin biosynthesis. Key gene families related to lignin synthesis, such as *cinnamoyl-CoA reductase*, *caffeoyl-CoA 3-O-methyltransferase*, *cinnamate 4-hydroxylase* and *4-coumarate: CoA ligase*, expanded and were highly expressed in the xylem, which may explain its excellent wood properties. Certain structural variations in lignin biosynthesis-related genes have been associated with genes identified in previous studies, potentially contributing to lignin biosynthesis through these variations and meriting further investigation. This study integrates genome annotation, comparative genomics, and transcriptomics, laying a foundation for functional analysis and molecular breeding for important characteristics of *E. cloeziana* and thus promoting the cultivation of new varieties of fast-growing trees.

1. Introduction

Eucalyptus, a collective term for species within the genera *Eucalyptus*, *Angophora*, and *Corymbia*, which are all members of the Myrtaceae family, is native to Australia and has a natural range extending throughout tropical and subtropical regions. Characterized by its rapid growth and short rotational periods, *Eucalyptus* is recognized as one of the three most important fast-growing plantation tree genera worldwide (Grattapaglia et al., 2012). Its extensive applications in pulp production, wood-based panels, and furniture manufacturing have resulted in significant economic and social benefits (Lee et al., 2024). In China, *Eucalyptus* has been cultivated for more than 130 years, with current plantations covering 5.46 million hectares (6.87 % of the nation's total plantation area) and yielding more than 30 million cubic meters of timber annually, accounting for 38 % of national wood production. This robust production system plays a pivotal role in alleviating timber

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supply–demand imbalances and advancing forestry industrialization (He et al., 2022). Among *Eucalyptus* species, *E. cloeziana* is distinguished by its high-quality wood, which exhibits color and texture similarities to those of *Dalbergia* species, earning it the moniker "Australian Dalbergia". Despite its slower growth rate than other *Eucalyptus* species, *E. cloeziana* has substantial research value and economic potential in premium timber markets (Li et al., 2012; Huang et al., 2018; Lan et al., 2022). Current research has primarily focused on germplasm introduction, with limited exploration of growth and development regulation and wood formation mechanisms (Alcorn et al., 2012; Listyanto et al., 2013; Zhou et al., 2014).

In plants, secondary growth primarily results from vascular cambium activity, which coordinates proliferative activity of cambium, differentiation and expansion of xylem cells, secondary cell wall (SCW) biosynthesis, and programmed cell death (Zhang et al., 2014). Among these factors, cell wall structural remodeling and chemical modification critically govern the physicochemical properties of wood. Lignification, a pivotal step in SCW development, facilitates the spatial deposition of secondary metabolites, resulting in the functional specialization of cell walls (Meents et al., 2018). Three architecturally distinct strata constitute the plant cell wall: the middle lamella, primary wall, and SCW (Zhang et al., 2021). The highly specialized structure and composition of the SCW make it the core contributor to wood properties, and elucidating the molecular mechanisms underlying wood formation mainly involves studying SCW biosynthesis and regulation (Fry, 2011).

Lignin, cellulose, and hemicellulose constitute the main components of the SCW, providing the fundamental framework of plant cell walls, and maintaining their structural integrity and mechanical strength (Meents et al., 2018), and variations in the relative abundance of these components significantly affect the mechanical performance of wood cell walls (Wrobel-Kwiatkowska et al., 2007). A reduction in the lignin and hemicellulose contents decreases cell wall rigidity, with lignin, a highly cross-linked aromatic polymer, having a greater impact on hardness than hemicellulose does (Gindl et al., 2002). Determining the gene regulatory networks and epigenetic modifications involved in SCW biosynthesis has become a central focus in wood formation research and has significant theoretical implications for the genetic improvement and development of functional biomaterials from trees.

Advances in high-throughput sequencing have revolutionized plant genomics studies. The first reference genome for *E. grandis* became available in 2014, significantly propelling research in *Eucalyptus* genomics and population genetics (Myburg et al., 2014). Advances in third-generation sequencing coupled with Hi-C chromatin interaction have substantially increased the contiguity and accuracy of genome assemblies. In 2023, researchers deployed an integrated multi-platform approach through leveraging Illumina short-reads, PacBio HiFi, and Hi-C technologies to produce a chromosome-scale hybrid genome assembly for *E. urophylla* × *E. grandis* (Shen et al., 2023). Recently, Australian National University released an updated assembly for *E. grandis* (GCF_016545825.1) that demonstrated greater contiguity than previous versions did (Ferguson et al., 2024). The ongoing development of *Eucalyptus* genomics has generated abundant resources for comparative genomics analyses, which have uncovered conserved and species-specific genetic features, including homologous gene sequences, unique/shared genomic elements, synteny relationships, phylogenetic frameworks, and evolutionary trajectories through cross-species genome sequence comparisons. These insights are critical for understanding adaptive evolution and species diversification (Grueber, 2015). Despite these advancements, genomics studies on *E. cloeziana* remain limited, with most research limited to phenotypic assessments of wood properties (Marini et al., 2022).

This research established a comprehensive genomic resource for *E. cloeziana* through high-quality genome annotation and transcriptome profiling, integrated with cross-species comparative analyses involving *E. grandis* and its hybrid *E. urophylla* × *E. grandis*. This approach aimed to explore the phylogenetic relationships among the *Eucalyptus* genus and to offer a genetic information for investigating the molecular pathways involved in wood formation. Our findings provide valuable insights into tree functional genomics and offer theoretical support for molecular breeding and the development of high-performance wood materials.

2. Materials and methods

2.1. Plant materials

Six-year-old healthy *E. cloeziana* individuals cultivated in a provenance trial plantation at Dongmen Forest Farm, Guangxi, China (107.84°E, 22.34°N), were selected as experimental materials in our study. Sample collection was conducted in January 2023. The region is characterized by a subtropical continental monsoon climate, with an average annual temperature of 22 °C and an average annual precipitation of approximately 1250 mm. Summers are hot and rainy, with average temperatures ranging from 25 °C to 30 °C, while winters are mild with low precipitation, with average temperatures between 10 °C and 15 °C. The planting site is flat, situated at an elevation of about 170 m, with red soil derived from sandstone bedrock. Tissues, including apical buds from the main stem, leaves, roots, xylem, and phloem, were harvested from each tree, and three biological samples were prepared per tissue type, resulting in 15 samples. To collect phloem and xylem samples, a compass was first used to determine the sampling direction, and marks were made on the north and south sides of each selected tree. Then, at a height of 1.3 m above the ground, an increment borer was used to drill vertically into the trunk along the north-south axis to obtain a complete core sample passing through the pith. The collected core sample was then sequentially divided into phloem and xylem sections, which were carefully cut into corresponding segments using a single-edged razor blade and separately stored for subsequent experimental analysis. Upon collection completion, all samples underwent immediate snap-freezing in liquid nitrogen prior to long-term storage at −80°C for RNA preservation, and ensuring the reliability of the transcriptomic analysis.

2.2. SMRT and Illumina sequencing

Total RNA of the 15 collected samples was extracted with the Tiangen Plant Kit (Beijing, China), after which, 15 Illumina RNA sequencing (RNA-Seq) libraries and one pooled library for single-molecule real-time (SMRT) sequencing were constructed. Biomarker Technologies Corporation (Beijing, China) executed all sequencing workflows. Subsequent bioinformatics analysis was conducted on a Dell Precision T7920 workstation in our laboratory.

2.3. Iso-Seq data analysis

Isoform sequencing (Iso-Seq) datasets were analyzed via PacBio's Iso-Seq v3.4.0 workflow (Gordon et al., 2015). Circular consensus sequencing (CCS) reads were generated from raw subreads using CCS v6.2.0 (Wenger et al., 2019) to ensure high base accuracy and to provide a reliable foundation for downstream analyses. Primer-specific recognition and removal were performed using Lima v2.1.0 (https://github.com/lima-vm/lima). The poly(A) tails and concatemer sequences were trimmed using the Iso-Seq refine module. Full-length non-chimeric (FLNC) reads were then subjected to iterative clustering and error correction with iterative clustering for error correction (ICE) to generate high-quality consensus isoforms. The ICE algorithm significantly enhanced sequence accuracy and consistency through multiple rounds of clustering and correction. By Minimap2 v2.24 (Li, 2018), the polished consensus isoforms were aligned to the genome of *E. cloeziana* (GCA_014182715.1).

2.4. Gene identification and functional annotation

In accordance with the protocol of Shen et al. (2023), genome annotation of *E. cloeziana* was performed. A species-specific repeat library was built through RepeatModeler v2.0.3 (https://www.repeatmasker.org/) and combined with the RepBase v21.12 (Bao et al., 2015) database to comprehensively identify repetitive elements across the genome using RepeatMasker v4.1.3 (Flynn et al., 2020). Four tools were used for *de novo* gene prediction: GeneMark-ET v4.38 (Borodovsky and McIninch, 1993), SNAP v7.0 (Korf, 2004), AUGUSTUS v3.4.0 (Stanke et al., 2006), and BRAKER v2.1.6 (Bruna et al., 2021). Homology-based predictions were performed with GenomeThreader v1.7.3 (Myburg et al., 2014) by aligning protein sequences from *A. thaliana*, *O. sativa*, *P. trichocarpa*, and *E. grandis*. For transcriptome-based evidence, StringTie v2.2.1 (Pertea et al., 2015) was utilized for the *de novo* assembly of Illumina RNA-Seq data. Meanwhile, Trinity v2.8.5 (Haas et al., 2003) was employed for genome-guided transcriptome assembly. Iso-Seq data were integrated using PASA v2.5.2 (Haas et al., 2008) to construct a comprehensive transcriptome database. TransDecoder v5.5.0 (Haas et al., 2013) was then applied to predict the coding sequences from the assembled transcripts. All evidence types—*de novo* predictions, homology-based annotations, and transcript-based evidence—were integrated by MAKER3 v3.1.3 (Cantarel et al., 2008) to optimize gene model prediction. EVM v2.1.1 (Haas et al., 2008) was subsequently used to generate weighted consensus gene models. A final round of annotation updates was performed with PASA v2.5.2 (Haas et al., 2008) to refine gene structures and predict alternative splicing events as well as 5′ and 3′ untranslated regions. Benchmarking Universal Single-Copy Orthologs (BUSCO) v5.2.2 (Simao et al., 2015) was employed against the relevant lineage-specific database to assess the completeness of the annotated gene set. The functional annotation of the genes was performed by querying multiple databases, including Swiss-Prot (Apweiler et al., 2004), NCBI-NR (Pruitt et al., 2007), Pfam (Mistry et al., 2021), eggNOG (Cantalapiedra et al., 2021), Gene Ontology (GO) (Ashburner et al., 2000), Kyoto Encyclopedia of Genes and Genomes (KEGG) (Kanehisa et al., 2004), and euKaryotic Orthologous Groups (KOG) (Koonin et al., 2004), to assign biological functions to the *E. cloeziana* gene models. A Circos plot of the whole genome of *E. cloeziana* was constructed by Circos v0.69.9 (Krzywinski et al., 2009).

2.5. Non-coding RNA prediction

The annotations of non-coding RNAs (ncRNAs) in *E. cloeziana* included long non-coding RNAs (lncRNAs), transfer RNAs (tRNAs), small nuclear RNAs (snRNAs), microRNAs (miRNAs), and ribosomal RNAs (rRNAs). Candidate lncRNAs were identified based on full-length transcript assemblies generated through the Iso-Seq data. Transcripts with a length $\geq$ 200 nucleotides and containing at least two exons were initially chosen as lncRNA. These were further filtered using multiple computational tools with coding potential assessment capabilities, including CNCI v2.0 (Sun et al., 2013), CPC2 v0.1 (Kang et al., 2017), PLEK v2 (Li et al., 2024), and Pfam v35.0 (Mistry et al., 2021), to reliably distinguish non-coding transcripts from protein-coding transcripts. tRNAs were predicted using tRNAscan-SE v2.0.12 (Lowe and Eddy, 1997). snRNAs and miRNAs were identified by aligning sequences against the Rfam database (Kalvari et al., 2018) through Infernal v1.1.4 (Nawrocki and Eddy, 2013). rRNAs were annotated using BLAST v2.14.0 (Kent, 2002) by aligning the genome against known rRNA sequences to accurately locate rRNA regions in the *E. cloeziana* genome.

2.6. Annotation of the chloroplast genome

The chloroplast genome of *E. cloeziana* (NCBI: CM024645.1) was systematically annotated using CPGAVAS2 (Shi et al., 2019) to provide a comprehensive identification of coding genes, rRNAs, and tRNAs. A circular map of the *E. cloeziana* chloroplast genome was produced utilizing Chloroplot (Zheng et al., 2020) to visualize its genomic structure and gene distribution. Simple sequence repeats were identified with MISA v2.1 (Beier et al., 2017). Tandem Repeats Finder (TRF) v4.10.0 (Benson, 1999) was used to detect long tandem repeats. Interspersed repeat elements were annotated using VMATCH v2.3.0 (http://www.vmatch.de). To investigate structural variation and evolutionary relationships, the boundaries of the IRb/SSC/IRa regions between chloroplast genomes of *E. cloeziana* and *E. grandis* were contrasted using CPJSDRAW (Li et al., 2023).

2.7. RNA-Seq analysis and identification of tissue-specific genes

Raw Illumina sequencing reads were subjected to quality control and filtering via fastp v0.23.2 (Chen et al., 2018). Clean data were then aligned to the *E. cloeziana* genome (GCA_014182715.1) by HISAT2 v2.0.5 (Kim et al., 2019). Gene abundance was quantified with StringTie v2.2.1 (Pertea et al., 2015), providing the basis for subsequent differential expression analysis. DESeq2 v1.48.1 package (Love et al., 2014) was employed to detect differentially expressed genes in R v4.4.0, with thresholds of $|\log_2(\text{fold change})| \geq 2$ and adjusted p-value (padj) < 0.05. Tissue-specific genes (TSGs) were determined in accordance with the criteria described by Wang et al. (2019): the expression level in the target tissue must be ≥ 1 and at least four times greater than that in any other tissue. In order to verify the RNA-Seq results, quantitative real-time PCR (qRT-PCR) was carried out on the chosen genes, ensuring the reliability and consistency of the transcriptomic data.

2.8. Phylogenetic analysis, divergence time estimation, and gene family analysis

Orthologous gene families were identified in 12 species via OrthoFinder2 v2.5.4 (Emms and Kelly, 2019). The divergence times were estimated using the mcmctree module in the PAML v4.10.6 (Yang, 2007), with calibration points obtained from the TimeTree v5 (http://www.timetree.org/) to ensure the temporal accuracy of phylogenetic inference. Gene family expansion and contraction events were subsequently employed by CAFE v4.2. The phylogenetic tree encompassing the genomes of the selected species was visualized with iTOL v5 (Letunic and Bork, 2021), enabling an intuitive presentation of evolutionary relationships.

2.9. Collinearity analysis and gene duplication detection

We integrated collinearity analysis with molecular evolutionary metrics to systematically elucidate the evolutionary history of *Eucalyptus* genomes. Whole-genome synteny comparisons were conducted between *E. grandis* and *E. cloeziana* using MCScanX v1.0.0 (Wang et al., 2012). Ks was estimated using the dmd and ksd modules in the wgd toolkit (Zwaenepoel et al., 2019). DupGen_finder v1.0.0 was executed using the default settings to detect diverse types of duplication events in the *E. cloeziana* genome (Qiao et al., 2019).

2.10. Structural variant identification

To identify structural variations (SVs) between *E. grandis* and *E. cloeziana*, we performed whole-genome alignment using Mummer v4.0.0 (Kurtz et al., 2004). The initial alignments were filtered to retain high-confidence regions. Structural variants were then detected using SyRI v1.5.4 (Goel et al., 2019), a specialized pipeline for genomic rearrangement analysis. Finally, large-scale SVs were visualized using Circos v0.69.9 (Krzywinski et al., 2009) to illustrate genome-wide divergence patterns.

3. Results

3.1. RNA-Seq and Iso-Seq analysis of E. cloeziana

To obtain a more comprehensive transcriptomic profile of *E. cloeziana*, RNA-Seq was performed on bud, leaf, root, phloem, and xylem samples. The sequencing data were high quality, with Q30 values exceeding 90 % across all samples, and the percentage of the clean reads mapped to the genome of *E. cloeziana* ranged from 84.89 % to 93.62 % (Table S1). High-quality RNA-Seq data provide a robust foundation for subsequent genome annotation and gene expression profiling.

This study employed SMRT to perform Iso-Seq on mixed tissue samples of buds, leaves, roots, phloem, and xylem to provide more accurate and comprehensive full-length transcripts for *E. cloeziana* and overcome the restrictions of conventional short-read sequencing technologies in terms of transcript completeness. Iso-Seq analysis generated 37,399,815 raw subreads, totaling 48.75 Gb. The use of PacBio's unique CCS mode significantly improved the sequencing accuracy, producing 951,650,885 bases of high-quality CCS reads with an average length of 1653 bp. After removing the poly(A) and concatemer sequences, 495,081 FLNC sequences were obtained, with an average length of 1397 bp. Using the ICE algorithm, 153,923 consensus isoforms were constructed, of which 99.97 % (153,872) were classified as high-quality isoforms. Only 51 low-quality transcripts were identified (Table S2). These results indicate that the full-length transcriptome data from this study are highly reliable, improving the transcriptome information for *E. cloeziana* and providing high-quality data for predicting the gene structure, analyzing alternative splicing, discovering new genes, and conducting comparative genomics.

3.2. Annotation of the nuclear and chloroplast genomes of E. cloeziana and comparison with genome assemblies and annotations of E. grandis

The full-genome Circos plot of *E. cloeziana* revealed that chromosomes ranged from 33.92 to 58.96 Mb in length, with gene densities varying from 0 to 23 genes and GC contents ranging from 34.67 % to 52.57 % per 100 kb (Fig. 1A). Repeat sequences of the genome of *E. cloeziana* were identified, accounting for 50.51 % of the genome. Among these repeats, long terminal repeat elements (LTRs) represented the largest proportion, comprising 20.43 % of *E. cloeziana* genome, followed by long interspersed nuclear elements (LINEs, 2.04 %), DNA transposons (1.02 %), and short interspersed nuclear elements (SINEs, 0.03 %) (Table S3).

A comparative analysis of genome assemblies and annotations was conducted between *E. grandis* (GCF_016545825.1) and *E. cloeziana* (GCA_014182715.1). *E. grandis* and *E. cloeziana* each had 11 chromosomes and a GC content of 39.5 %. *E. cloeziana* had a smaller genome (480.10 Mb) but fewer scaffolds than *E. grandis* (615.90 Mb) (Table 1). We assessed the previously published protein annotations of *E. cloeziana* (Ferguson et al., 2024) by utilizing BUSCO with the Viridiplantae_odb10 and Embryophyta_odb10 datasets, which yielded completeness scores of 85.4 and 84.4 %, respectively, with single-copy gene proportions of 81.9 and 80.1 %, respectively (Table S4).

In our study, we reannotated the *E. cloeziana* genome by integrating RNA-Seq and Iso-Seq data, which significant improved BUSCO completeness scores to 95.1 % (viridiplantae_odb10) and 92.7 % (embryophyta_odb10), with single-copy gene proportions exceeding 90 % for both datasets (Table S4). We further compared the number of genes, mRNAs, and ncRNAs annotated in the *E. grandis* and *E. cloeziana* genomes (Table 2). A sum of 41,610 protein-coding genes was annotated, generating 79,394 mRNA transcripts in the *E. cloeziana* genome. Although the number of genes was similar to that in *E. grandis* (42,619), *E. cloeziana* presented a greater number of mRNA isoforms, likely due to the incorporation of Iso-Seq data, which captured more extensive alternative splicing events. Among the 41,610 protein-coding genes, 32,827 were functionally annotated in seven databases: NR (32,530; 78.18 %), eggNOG (27,079; 65.08 %), GO (26,218; 63.01 %), Pfam (24,733; 59.44 %), Swiss-Prot (22,305; 53.60 %), KEGG (21,581; 51.86 %), and KOG (16,300; 39.17 %) (Table S5). Compared to the *E. grandis* genome, we annotated more types of ncRNAs in *E. cloeziana*. 1301 lncRNAs, 265 tRNAs, 226 snRNAs, and 174 miRNAs were newly predicted, and 123 rRNAs were annotated in *E. cloeziana* (Table 2). These rich annotation data offer key references for follow-up gene expression analysis and comparative genomics.

Based on the previously sequenced chloroplast genome of *E. cloeziana* (NCBI: CM024645.1), we carried out genome annotation and feature analysis. The chloroplast genome of *E. cloeziana* had a typical quadripartite structure with a length of 160,184 bp (Fig. 1B). We revealed that the chloroplast genome contained 80 protein-coding genes, 37 tRNAs, and 8 rRNAs (Fig. 1B, Table S6). *E. grandis* had two copies of each of the *rpl23* and *ycf1* genes, whereas *E. cloeziana* had one copy of each. In *E. grandis*, *rpl23* was distributed in both the IRb and IRa regions, whereas in *E. cloeziana,* it was only located in the IRb region. The *ycf1* gene spanned the JSA and JSB boundaries in *E. grandis* but spanned only the JSA boundaries in *E. cloeziana* (Fig. 1C). This comparative analysis of chloroplast genomes enhances comprehension of the structural dynamics and evolutionary divergence of *Eucalyptus* species, providing a better understanding of their phylogeny and taxonomy.

3.3. Characterization of mRNAs and lncRNAs in E. cloeziana

Based on improved genome annotation, this study further focused on lncRNA characterization and analyzed structural differences between lncRNAs and mRNAs. The 1301 lncRNAs identified in *E. cloeziana* using Iso-Seq were classified into four categories: sense lncRNAs (644, 49.50 %), intergenic lncRNAs (454, 34.90 %), antisense lncRNAs (155, 11.91 %), and intronic lncRNAs (48, 3.69 %) (Fig. 2A). Compared with mRNAs, lncRNAs generally contained fewer exons, with the majority comprising two exons (Fig. 2B). Transcript length analysis revealed that lncRNAs were primarily within the 0–5000 bp range, whereas mRNAs exhibited a broader distribution spanning 0–10,000 bp (Fig. 2C). The exon length distribution patterns of the lncRNAs and mRNAs were similar, with most exons being 0–1000 bp in length (Fig. 2D). These structural features highlight the distinct differences between lncRNAs and mRNAs, suggesting that lncRNAs play diverse roles in gene expression regulation.

3.4. Identification and characterization of tissue-specific gene (TSG) expression in E. cloeziana

This study systematically identified 3137 TSGs using RNA-Seq analysis of five tissue types from *E. cloeziana*. Among these, buds (1278) presented the greatest number of TSGs, followed by the xylem (609), leaves (536), phloem (477), and roots (237) (Fig. 3A). TSGs revealed the biological functions associated with each tissue type. The TSGs in the xylem were enriched mainly in the regulation of the actin cytoskeleton, phenylalanine metabolism, and flavonoid biosynthesis. Lignin biosynthesis begins with phenylalanine, which is transformed into cinnamic acid via the action of phenylalanine ammonia-lyase (Fig. S1). This process is followed by a series of enzyme-catalyzed reactions to produce lignin monomers (Hao and Mohnen, 2014), which are key components in the formation of SCW, indicating that certain genes related to lignin biosynthesis in vascular tissue make significant contributions to SCW biosynthesis.

qRT-PCR was carried out on ten TSGs to verify the reliability of the RNA-Seq data. These genes included the bud-specific regulatory genes *WUSCHEL* (*WUS*) and *CENTRORADIALIS* (*CEN*), the leaf photoprotection-related genes *state transition 7* (*STN7*) and *Photosystem II reaction center W* (*PsbW*), the root development-related genes *leucine-rich repeat* (*LRR*) and *Zinc finger*, the phloem transport regulatory genes *CLAVATA3/ESR-Related* (*CLE*) and *walls are thin 1* (*WAT1*), and key

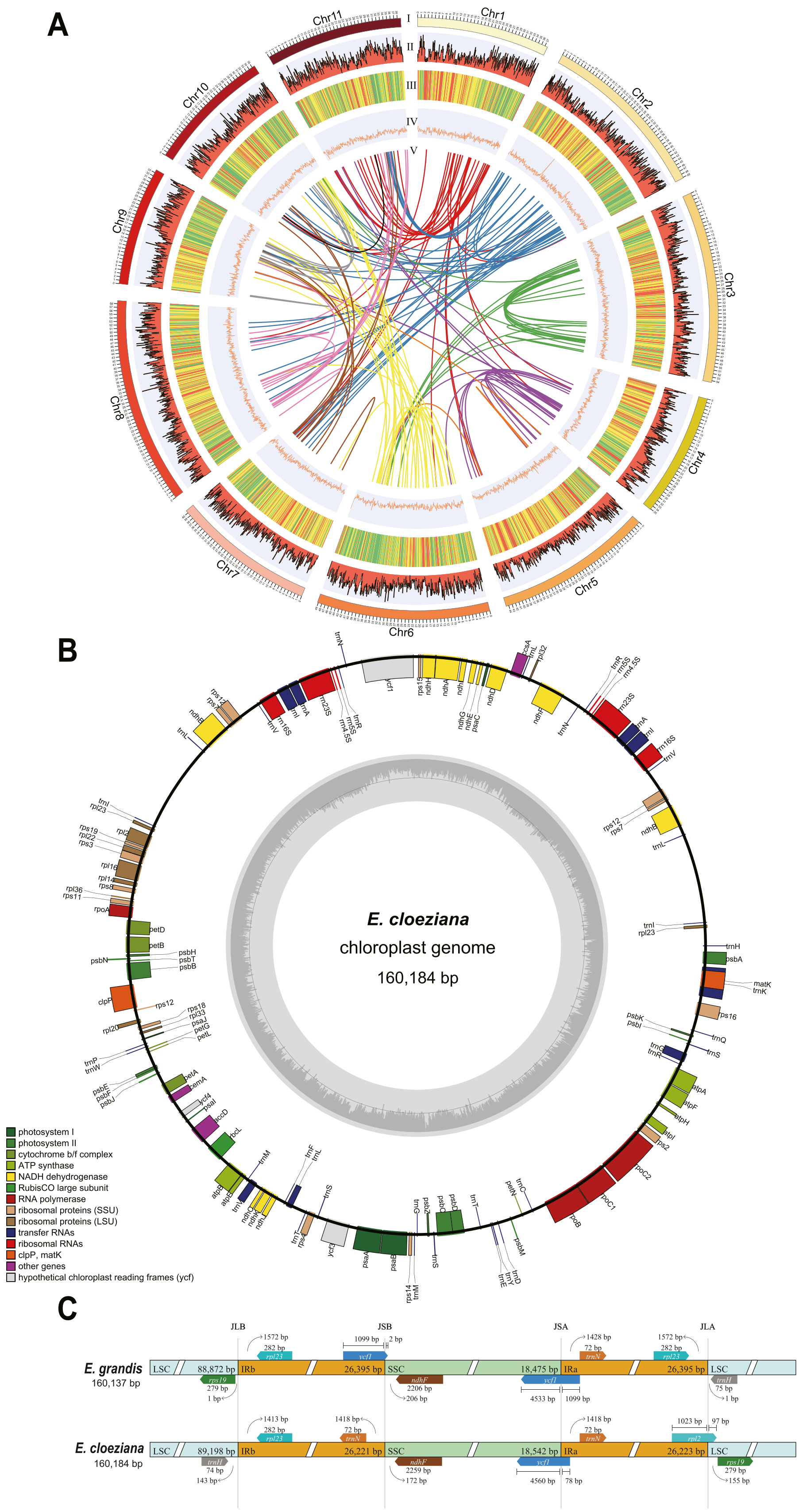

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Fig. 1. Characteristics of the chromosome and chloroplast genomes of *Eucalyptus cloeziana* (A) Genomic characteristics of *E. cloeziana* chromosomes. I: The 11 pseudomolecules of *E. cloeziana*; II: gene density distribution; III: heatmap of transposable elements (TEs); IV: GC content distribution; V: syntenic blocks in the *E. cloeziana* genome, represented by colored links. All tracks are visualized in a 100-kb window. (B) Annotation map of the *E. cloeziana* chloroplast genome. The circular map illustrates the gene organization in the plastome. Genes are assigned distinct colors based on their function. (C) Comparative analysis of LSC, SSC, and IR region boundaries in the chloroplast genomes of *E. grandis* and *E. cloeziana.*

Table 1
Statistics of the genome assemblies of *Eucalyptus grandis* and *E. cloeziana*.

Genomic feature	*E. grandis* (Ferguson et al., 2024)	*E. cloeziana* (Ferguson et al., 2024)
Genome size (Mb)	615.90	480.10
Assembly level	chromosome	chromosome
Number of chromosomes	11	11
Number of scaffolds	35	24
Scaffold N50 (Mb)	58.50	44.70
Scaffold L50	5	5
GC content	39.50 %	39.50 %

Table 2
Comparison of genome annotations of *Eucalyptus grandis* and *E. cloeziana*.

Parameter	*E. grandis*	*E. cloeziana* (Ferguson et al., 2024)	*E. cloeziana* (in this study)
Number of genes	42,619	43,093	41,610
Number of mRNAs	44,989	43,093	79,394
Number of lncRNAs	5270	—	1301
Number of tRNAs	796	—	265
Number of snRNAs	134	—	226
Number of miRNAs	—	—	174
Number of rRNAs	556	—	123

xylem formation genes *phenylalanine ammonia-lyase* (*PAL*) and *irregular xylem* (*IRX*) (Fig. 3B–K). The *PAL* gene, which participates in lignin biosynthesis, and the *IRX* gene, which is involved in hemicellulose synthesis, are involved in SCW biosynthesis pathways (Meents et al., 2018). A strong positive correlation was identified through the correlation analysis between RNA-Seq and qRT-PCR expression levels ($R^2 = 0.79$, $P < 2.2e{-}16$) (Fig. 3L), confirming the accuracy and reliability of the transcriptomic data.

3.5. Genomic evolution of E. cloeziana and the identification of species-specific gene families through comparative genomics analysis

In this research, we carried out a thorough identification and comparative analysis of orthologous gene families on the basis of the genome data of *E. cloeziana, E. grandis*, *E. urophylla* × *E. grandis*, and nine representative species (*Oryza sativa*, *Arabidopsis thaliana*, *Solanum lycopersicum*, *Vitis vinifera*, *Salix suchowensis*, *Populus trichocarpa*, *Prunus persica*, *Juglans regia*, and *Quercus rubra*). We revealed lineage-specific genomic changes during evolution by constructing a species phylogenetic tree, estimating divergence times, and performing gene family expansion/contraction analysis, providing important insights into species relationships and evolutionary history. The phylogenetic tree created using single-copy orthologs of 12 species revealed that *E. grandis* and *E. urophylla* × *E. grandis* formed a monophyletic group, with *E. cloeziana* diverging from the common ancestor of this branch approximately 15.006 million years ago (Mya) (Fig. 4A).

To further validate the phylogenetic relationships among species, we constructed a phylogenetic tree according to the chloroplast genome, and the evolutionary relationships were consistent with those derived from a nuclear gene-based phylogeny (Fig. 4B). Analysis of comparative gene family evolution indicated that compared to the most recent common ancestor, *E. cloeziana* experienced an expansion in 1002 gene families and a contraction in 755 gene families (Fig. 4A). Further analysis of genome duplication events revealed that gene family expansion in *E. cloeziana* was mainly driven by dispersed duplication (DSD) events (69.46 %) (Fig. 4C).

Among the gene families in *E. cloeziana*, 8885 were in common with the other 11 species, and 1224 were identified as species-specific gene families (Fig. 4D). We performed functional enrichment analysis on the species-specific gene families of *E. cloeziana*. GO enrichment analysis mainly focused on glycosyltransferase and hexosyltransferase activities and UDP-glycosyltransferase activity (Fig. 4E). KEGG pathway analysis revealed the following pathways: the MAPK signaling pathway—plant, plant hormone signal transduction, and protein processing in the endoplasmic reticulum (Fig. 4F). Synteny analysis revealed that 1,828 collinear blocks between the genomes of *E. cloeziana* and *E. grandis,* which were located primarily on orthologous chromosomes, indicating highly conserved and contiguous collinearity at the chromosomal level and confirming genomic synteny and diploid genome conservation (Fig. 5A and B).

We systematically identified duplication events throughout the *E. cloeziana* genome, and the expansion of its genome size was primarily driven by DSD and whole-genome duplication (WGD) events (Table S7). WGD is a key mechanism driving plant genome expansion, and it has been widely reported among various species (Ren et al., 2018). We employed synonymous substitution (Ks) and fourfold synonymous (degenerative) third-codon transversion (4DTv) analyses to identify WGD events. The concordant trends observed in the Ks and 4DTv curves showed that *E. grandis* and *E. cloeziana* each underwent independent WGD events after speciation (Fig. 5C and D). This finding further supports the occurrence of independent WGD events in both species after divergence, driving the evolution of their respective genomes. Gene family expansion may directly influence the expansion in the copy numbers of genes implicated in secondary metabolic pathways, providing an evolutionary explanation for subsequent investigations into the molecular basis of SCW synthesis in *E. cloeziana*.

3.6. Genome-wide landscape of structural variations (SVs) between E. grandis and E. cloeziana

By comparing the genomes of *E. cloeziana* and *E. grandis* genomes, we identified 15,201 SVs, comprising 5646 translocations (TRANS; 37.14 %), 5102 duplications (DUP; 33.56 %), 1421 deletions (DEL; 9.35 %), 1384 insertions (INS; 9.10 %), 797 copy losses (CPL; 5.24 %), 717 copy gains (CPG; 4.72 %) and 134 inversions (INV; 0.88 %) (Fig. S2 A). Chromosome-by-chromosome alignment of the *E. cloeziana* and *E. grandis* genomes, we observe an overall high degree of collinearity interleaved with distinct structural rearrangements including TRANS, DUP and INV (Fig. S3). This genome-wide landscape highlights both conserved architecture and species-specific SVs that may underlie genome divergence between the two *Eucalyptus* species. Annotation of SVs revealed that most SVs fall in exonic (34.26 %) and intergenic regions (33.67 %) (Fig. S2 B). In order to clarify the distribution characteristics of different types of genome variation at the chromosome level, the distribution density of different variations is displayed by genome Circos diagram which revealed distinct distribution patterns of SVs across *E. cloeziana*'s 11 chromosomes, with TRANS and DUP showing widespread distribution while INV was the least abundant (Fig. S2 E). GO analysis and KEGG pathway (Fig. S2 C and D) revealed terpenoid biosynthesis (sesquiterpenoid/diterpenoid), cell wall metabolism (cutin/wax biosynthesis, primary cell septum), and stress adaptation (brassinosteroid signaling, cellular response to amino acid stimulus, calcium channel activity). Moreover, these findings suggest that genes involved in lignin biosynthesis in *E. cloeziana* are frequently affected by

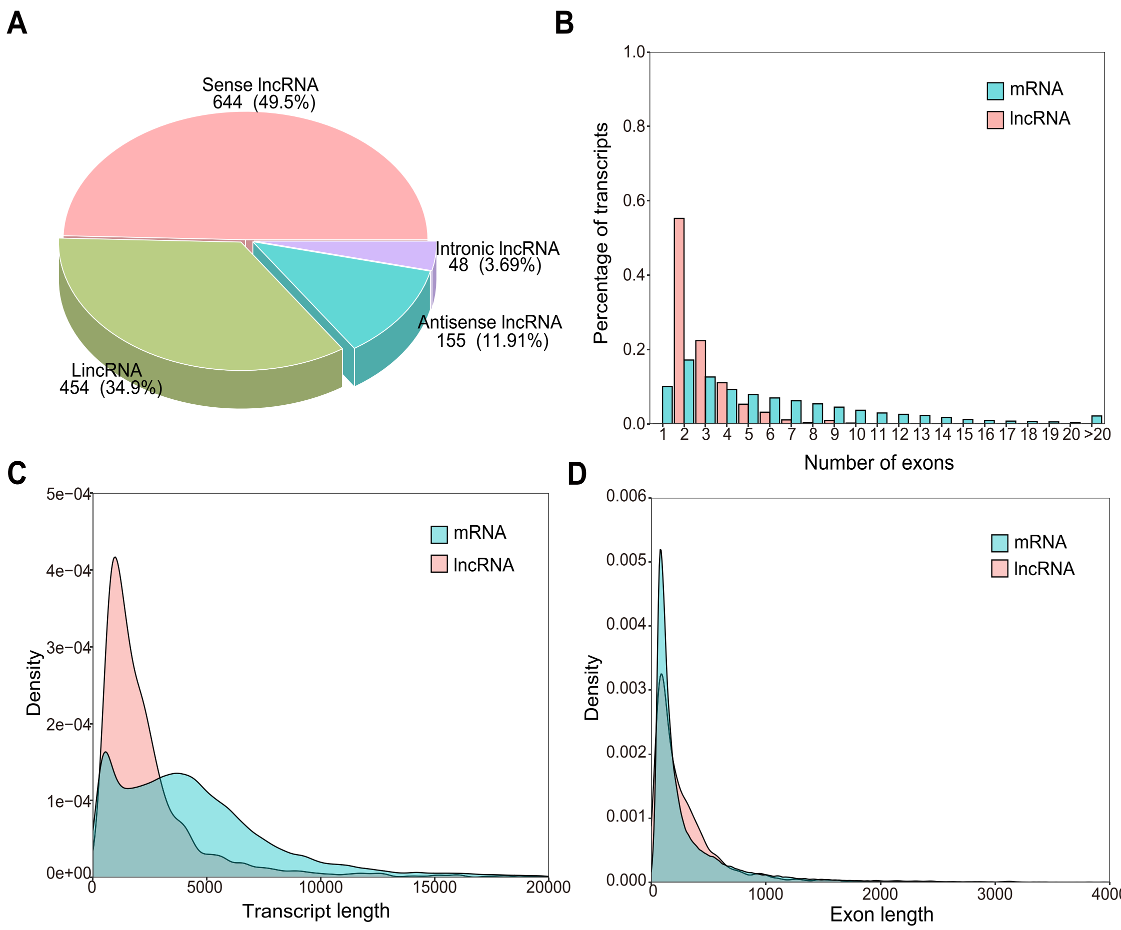

Fig. 2. Characteristics of *Eucalyptus cloeziana* mRNAs and lncRNAs (A) Classification of lncRNAs. (B) Percentage of exons in mRNAs and lncRNAs. (C) Comparison of the transcript lengths of mRNAs and lncRNAs. (D) Density distribution of the exon lengths of mRNAs and lncRNAs.

structural variations, primarily TRANS, followed by DUP and INV, potentially contributing to wood property diversity and adaptive evolution (Table S8).

3.7. Expression of genes related to lignin, cellulose, and hemicellulose biosynthesis in E. cloeziana

Comparative genomics analysis of *E. cloeziana* and *E. grandis* identified 1275 *E. cloeziana*-specific genes (Fig. 6A). In the KEGG pathway analysis, these genes displayed significant enrichment in the MAPK signaling pathway—plant, starch and sucrose metabolism (including cellulose biosynthesis), and phenylpropanoid biosynthesis (involving lignin biosynthesis) (Fig. 6B). This study focused on the biosynthesis pathways of plant SCW and systematically analyzed the transcriptional regulation characteristics of lignin, cellulose, and hemicellulose in *E. cloeziana*. The genes associated with the biosynthesis of these three substances exhibited significant tissue-specific expression patterns across the five examined tissues. By annotating gene functions and comparing homologous genes, ten gene families related to lignin biosynthesis were identified in *E. cloeziana*, comprising 122 members, which is approximately 1.13 times greater than the number in *E. grandis*. Notably, the *cinnamoyl-CoA reductase* (*CCR*), *caffeoyl-CoA 3-O-methyltransferase* (*CCoAOMT*), and *4-coumarate: CoA ligase* (*4CL*) families showed remarkable expansion, with higher gene copy numbers than *E. grandis* did (Fig. 6C).

For the cellulose and hemicellulose biosynthesis pathways, 11 gene families with 114 associated genes were identified in *E. cloeziana*, among which the number of members of the *cellulose synthase* (*CESA*) gene family significantly increased (Fig. 6D). Based on the metabolic pathways of lignin, cellulose, and hemicellulose reported in the literature (Boerjan et al., 2003; Myburg et al., 2014; Carocha et al., 2015; Polko and Kieber, 2019) and the transcriptome data, the gene expression across gene family members was normalized, and a metabolic pathway heatmap was generated. Key candidate genes for lignin biosynthesis such as *PAL*, *cinnamate 4-hydroxylase* (*C4H*), *hydroxycinnamoyl-CoA: shikimate/quinate hydroxycinnamoyl transferase* (*HCT*), and *CCR* presented high expression levels in xylem tissue (Fig. 6E). Furthermore, in the cellulose and hemicellulose biosynthesis pathways, the expression levels of genes other than the *UDP-xylose synthase* (*UXS*) and *UDP-glucose pyrophosphorylase* (*UGP*) were substantially higher in the xylem tissue compared to other tissues (Fig. 6F). Through comparative genomics and transcriptomic expression profile analysis, this study employed a multi-omics integration strategy to reveal the expansion trends and tissue expression characteristics of gene families related to SCW biosynthesis in *E. cloeziana*, providing a theoretical basis and genetic resource to comprehend the molecular regulatory mechanisms of SCW formation in hardwood plants.

4. Discussion

4.1. Annotation of the E. cloeziana genome

Eucalyptus is renowned for its fast growth, strong adaptability, and high stress resistance and is recognized as one of the three main fast-growing afforestation trees recommended by the Food and Agriculture Organization of the United Nations (Kaur and Monga, 2021; Koutika et al., 2022). Genomic research on *E. grandis*, which laid a foundation for dedicated efforts to improve the genetics of *Eucalyptus* species, resulted

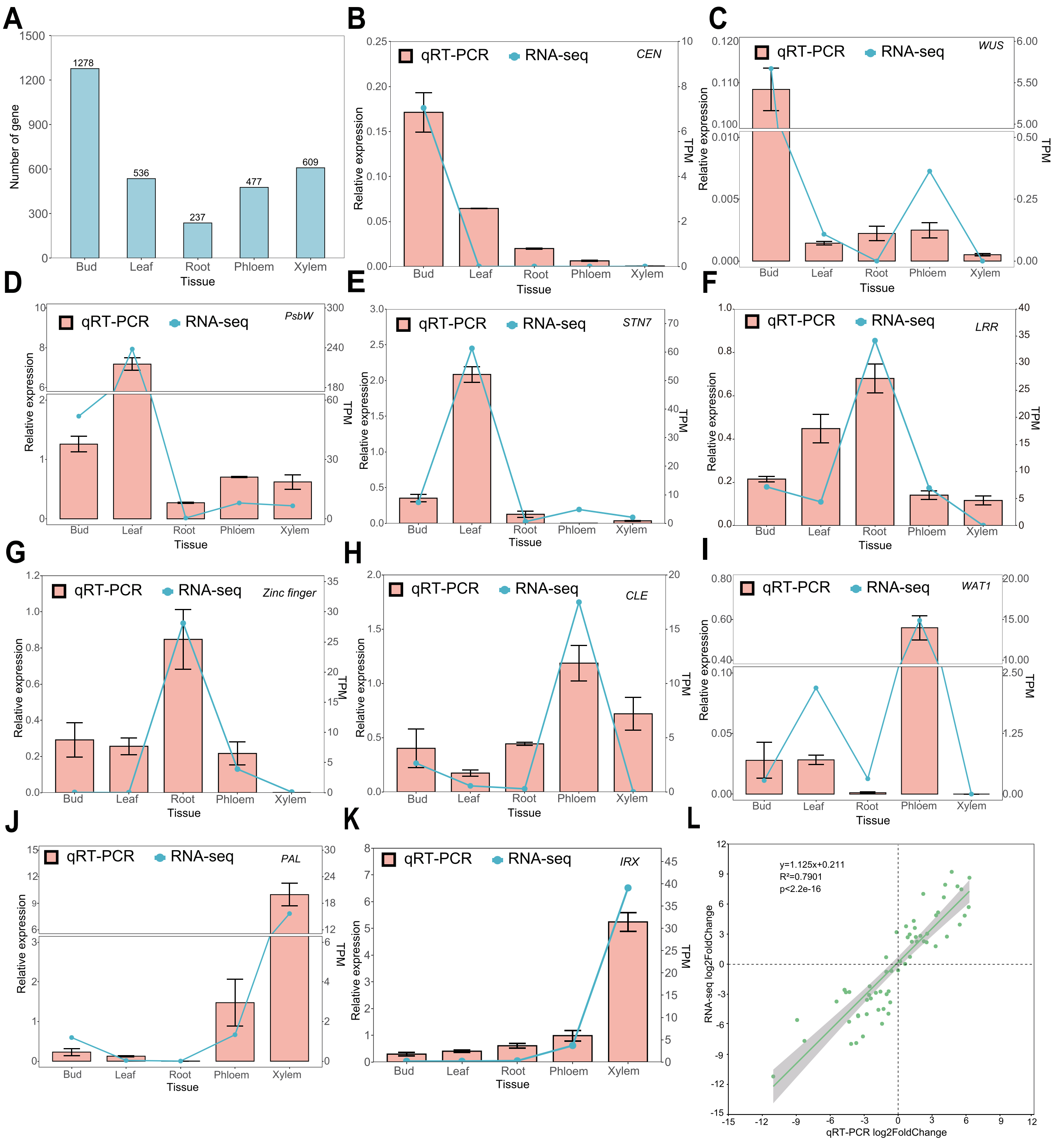

Fig. 3. Tissue-specific genes expression and qRT-PCR validation of RNA-Seq in *Eucalyptus cloeziana* (A) Number of tissue-specific genes expressed in *E. cloeziana*. (B–K) Analysis of relative expression levels of ten genes across five tissues, comparing the qRT-PCR results with the RNA-Seq results. The following genes are shown: *CEN* (*CENTRORADIALIS*), *WUS* (*WUSCHEL*), *PsbW* (*Photosystem II reaction center W*), *STN7* (*state transition 7*), *LRR* (*leucine-rich repeat*), *CLE* (*CLAVATA3/ESR-Related*), *WAT1* (*walls are thin 1*), *PAL* (*phenylalanine ammonia-lyase*), *IRX* (*irregular xylem*). (L) Correlation analysis between qRT-PCR and RNA-Seq data.

in the inaugural *E. grandis* genome assembly (640.00 Mb), which was assembled through the integration of multiple sequencing technologies and achieved 94 % sequence coverage, with 34 % consisting of tandem repeats (Myburg et al., 2014). An updated *E. grandis* genome assembly spans 615.90 Mb with a scaffold N50 of 58.50 Mb in the NCBI-GenBank, including the annotation of 42,619 protein-coding genes, thereby improving its utility as a reference genome (Ferguson et al., 2024). In 2023, the genome of *E. urophylla* × *E. grandis* was reported to span 545.80 Mb, with a scaffold N50 of 51.62 Mb and 34,502 genes annotated (Shen et al., 2023).

Eucalyptus cloeziana, a high-quality hardwood species, has promising potential for industrial and scientific research (Seng et al., 2022). A BUSCO assessment of the published *E. cloeziana* genome annotation (Ferguson et al., 2024) revealed a single-copy gene completeness of

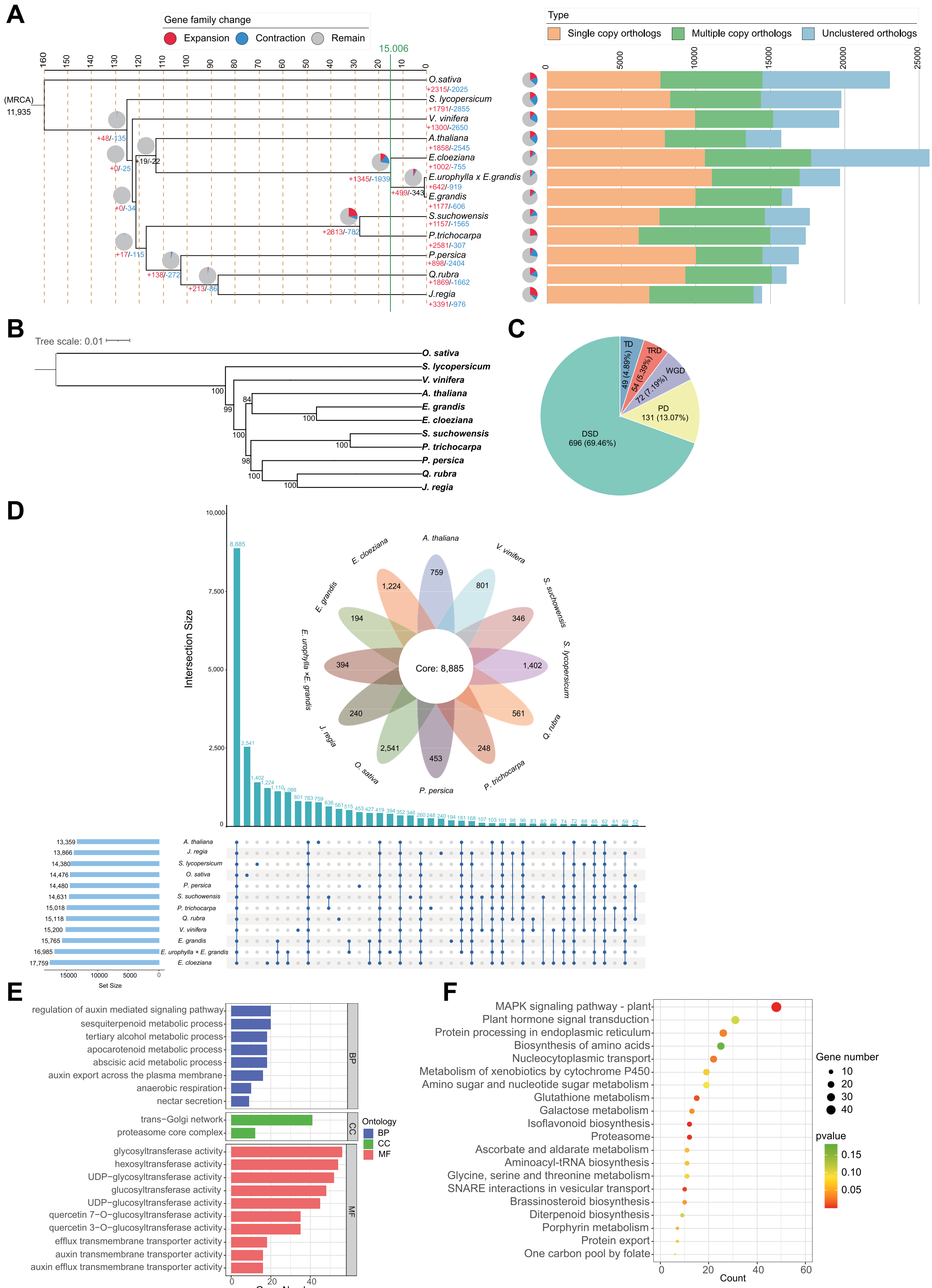

Fig. 4. Genome and gene family evolution in *Eucalyptus cloeziana* (A) Phylogenetic tree, divergence time estimation, and orthogroup analysis for *E. cloeziana* and 11 representative plant species. Numbers indicate gene family expansion (red) and contraction (blue). The bar plot represents the number of different types of orthologous gene families across the 12 species. (B) Phylogenetic tree based on chloroplast genomes. Bootstrap values for the main branches are also reported. (C) Duplication modes contributing to gene family expansion in *E. cloeziana*. The number of expanded gene families associated with each mode of duplication is shown. DSD: dispersed duplication, PD: proximal duplication; WGD: whole-genome duplication; TRD: transposed duplication; TD: tandem duplication. (D) Gene family cluster flower plot and UpSet diagram. In the flower plot, the center circle represents the number of gene families shared among all species, whereas each petal indicates the number of gene families unique to each species. The UpSet diagram displays the distribution of gene family clusters across species. (E) Top 20 GO enrichment terms for species-specific gene families in *E. cloeziana*. (F) Top 20 enriched KEGG pathways for species-specific gene families in *E. cloeziana*.

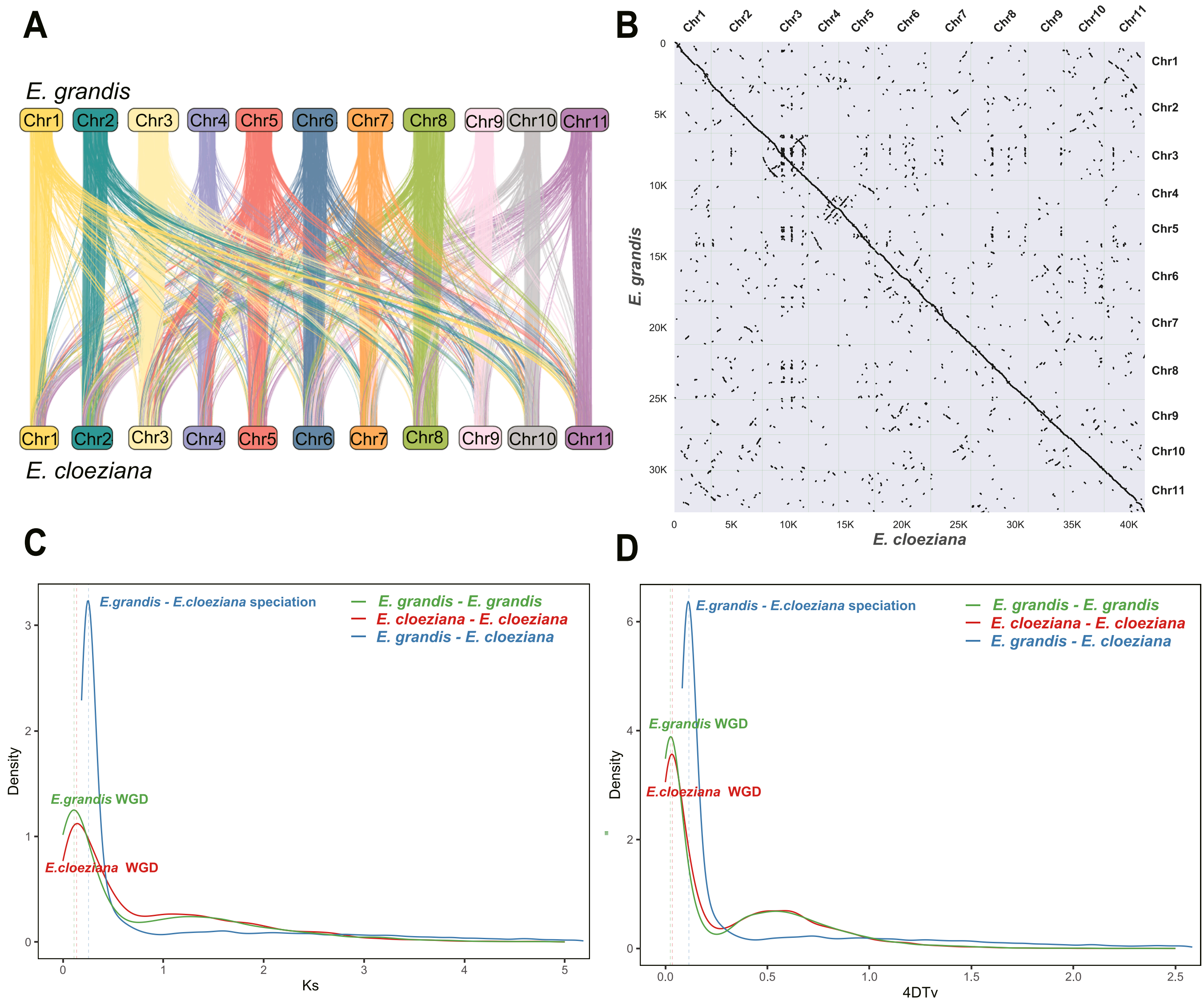

Fig. 5. Genome comparison analysis between *Eucalyptus grandis* and *E. cloeziana* (A) Genome collinearity analysis between *E. grandis* and *E. cloeziana*. (B) Dot plot of synteny blocks between *E. grandis* and *E. cloeziana*. (C) Synonymous substitution (Ks) distributions in *E. grandis* and *E. cloeziana*. (D) Fourfold synonymous (degenerative) third-codon transversion (4DTv) distributions in *E. grandis* and *E. cloeziana*.

80.1 % in the Embryophyta_odb10 database, which is likely attributed to the lack of *E. cloeziana* transcriptome data during the annotation process, which relies solely on homologous annotations from closely related species (Table S4). Therefore, in our study, RNA-Seq and Iso-Seq data from *E. cloeziana* were used to systematically analyze the genome annotation, identifying 41,610 genes. The optimized annotation achieved 90.0 % single-copy gene completeness in the Embryophyta_odb10 dataset, confirming the high quality of the reference genome annotation (Table S4). We also expanded the annotation of ncRNAs, including lncRNAs, tRNAs, snRNAs, miRNAs, and rRNAs, in *E. cloeziana*, which can be accessed through the EucaMOD database developed by our laboratory (http://eucalyptusggd.net/eucamod). This finding bridges an important gap in the study of ncRNA regulatory networks in *Eucalyptus*. Comparative analysis of the chloroplast genomes of *E. cloeziana* and *E. grandis* revealed evolutionary dynamics in the IR and LSC/SSC boundary regions (Fig. 1C), providing comprehension of the evolutionary mechanisms of *Eucalyptus* chloroplast genomes. Overall, this study enhances the genomic resources available for *Eucalyptus* and advances our understanding of hardwood genome evolution. The high-quality *E. cloeziana* genome provides a valuable molecular foundation for future studies on the comparative genomics and genetic improvement of *Eucalyptus* species.

4.2. Identification of TSGs in E. cloeziana

TSGs exhibit high expression levels in specific tissue types and confer distinct morphological characteristics and physiological functions to the corresponding cells. Previous studies have demonstrated clear organ-specific expression patterns for TSGs. For example, in *Populus euphratica*, 5310 TSGs presented a significant tissue distribution, with the highest number in leaves (2849), followed by phloem (1247), roots (679), and xylem (535) (Yu et al., 2017). In *Pinus sylvestris*, 603, 342, 220, 206, and 202 TSGs were identified in needles, buds, megagametophytes, embryos, and phloem, respectively (Cervantes et al., 2021). In this study, 3137 TSGs were identified in *E. cloeziana*, displaying a distinct distribution pattern: bud (1278), xylem (609), leaf (536), phloem (477), and root (237) (Fig. 3A). We performed KEGG enrichment analysis on the TSGs to investigate the tissue-specific regulation of metabolic pathways. Notably, xylem-specific genes were significantly enriched in pathways associated with the actin cytoskeleton, phenylalanine metabolism, and flavonoid biosynthesis (Fig. S1), highlighting that the xylem plays a critical role in cellular structural maintenance and

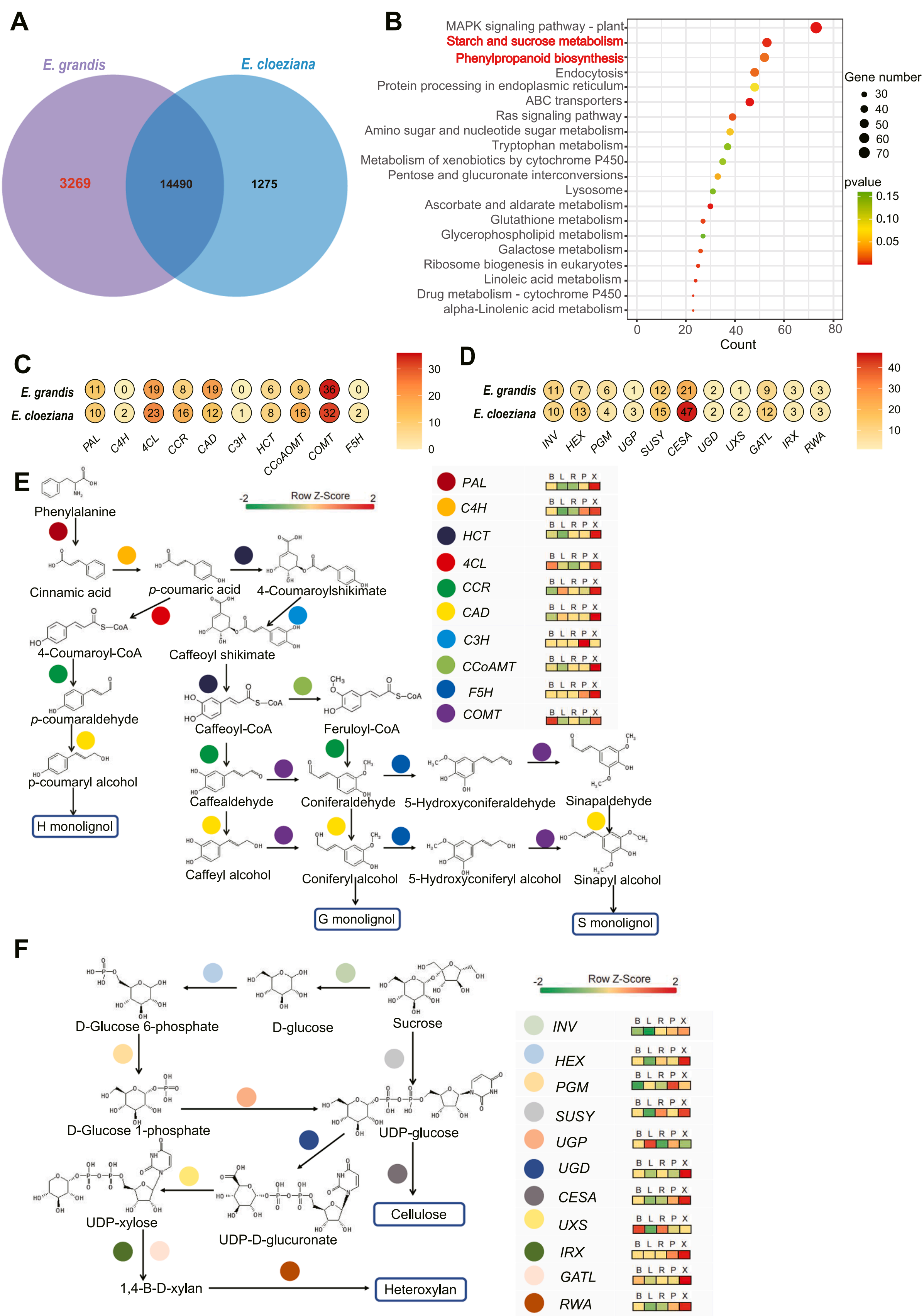

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Fig. 6. Species-specific gene families in *Eucalyptus cloeziana* and their involvement in KEGG enrichment and the biosynthetic metabolic pathways associated with lignin, cellulose, and hemicellulose (A) Venn diagram comparing gene families in *E. grandis and E. cloeziana*. (B) Top 20 enriched KEGG terms in *E. cloeziana*-specific gene families. (C) Number of genes involved in lignin biosynthesis pathways in *E. grandis and E. cloeziana*. (D) Number of genes involved in cellulose and hemicellulose biosynthesis pathways in *E. grandis and E. cloeziana*. (E) Expression of genes related to lignin biosynthesis in *E. cloeziana.* The following key genes involved in lignin biosynthesis are shown: *PAL* (*phenylalanine ammonia-lyase*), *C4H* (*cinnamate 4-hydroxylase*), *4CL* (*4-coumarate: CoA ligase*), *CCR* (*cinnamoyl-CoA reductase*), *CAD* (*cinnamyl alcohol dehydrogenase*), *C3H* (*4-coumarate 3-hydroxylase*), *HCT* (*hydroxycinnamoyl-CoA: shikimate/quinate hydroxycinnamoyl transferase*), *CCoAOMT* (*caffeoyl-CoA 3-O-methyltransferase*), *COMT* (*caffeic acid O-methyltransferase*), and *F5H* (*ferulate-5-hydroxylase*). (F) Expression of genes involved in cellulose and hemicellulose biosynthesis in *E. cloeziana.* The following key genes involved in cellulose and hemicellulose biosynthesis are shown: *INV* (*invertase*), *SUSY* (*sucrose synthase*), *HEX* (*hexokinase*), *PGM* (*phosphoglucomutase*), *UGP* (*UDP-glucose pyrophosphorylase*), *UGD* (*UDP-glucose dehydrogenase*), *CESA* (*cellulose synthase*), *UXS* (*UDP-xylose synthase*), *GATL* (*galacturonosyltransferase-like*), *IRX* (*irregular xylem*), and *RWA* (*reduced wall acetylation*).

secondary metabolite biosynthesis.

Lignin biosynthesis proceeds via the PAL-catalyzed deamination of phenylalanine to cinnamate, followed by a cascade of hydroxylation, methylation, and reduction reactions that generate monolignols that are ultimately polymerized into lignin (Hao and Mohnen, 2014). Compared to other tissues, PAL, a key enzyme in the phenylpropanoid pathway, was significantly more abundantly expressed within the xylem of *E. cloeziana*, implying that increased lignin precursor biosynthesis may participate in SCW thickening (Fig. 3J, Fig. 6E) (Pascual et al., 2016). The expression of *IRX* genes, which regulate enzymes and transcription factors involved in SCW biosynthesis and play essential roles in the formation of various cell wall polymers, was elevated in the xylem of *E. cloeziana* (Fig. 3K, Fig. 6F) (Turner and Somerville, 1997; Hao and Mohnen, 2014). Together, these findings provide perspectives on the functional landscape of TSG expression in *E. cloeziana* and offer valuable references for future studies on tissue development and SCW metabolic regulation in woody plants.

4.3. Comparative genomics reveals the evolution of E. cloeziana

Synteny analysis is a comparative genomic approach used to identify conserved gene orders and homologous relationships between the genomes of different species or within the same species, providing insights into genome organization and evolutionary dynamics (Tang et al., 2008). In this study, marked conserved synteny was found between the *E. grandis* and *E. cloeziana* genomes, detected with 1828 collinear blocks (Fig. 5A and B). Furthermore, phylogenetic analyses based on the nuclear and chloroplast genomes of *A. thaliana*, *E. grandis*, *E. cloeziana*, and other representative species revealed that *E. cloeziana* clustered in the same evolutionary clade as *E. grandis*. Notably, *E. urophylla* × *E. grandis* exhibited a closer phylogenetic affinity to *E. grandis* in the nuclear genome-based phylogeny (Fig. 4A and B), which is in accordance with earlier research (Shen et al., 2023). In contrast with *E. grandis* and *E. urophylla* × *E. grandis*, *E. cloeziana* showed a large number of gene family expansions and contractions, with expansions driven primarily by DSDs (Fig. 4A and C). Compared with those of *E. grandis*, the KEGG enrichment results of the *E. cloeziana*-specific genes (1275) revealed marked enrichment in pathways associated with starch and sucrose metabolism (including cellulose biosynthesis) and phenylpropanoid biosynthesis (involving lignin biosynthesis) (Fig. 6A and B). These results highlight the unique biosynthetic pathways associated with hardwood properties at the genomic level in *E. cloeziana*, providing a functional genomics perspective on the evolutionary divergence of *E. cloeziana* and other *Eucalyptus* species.

Previous studies have demonstrated a high degree of genomic collinearity between *E. grandis* and *E. urophylla*. Detailed comparative analyses have also revealed that their genomes harbor extensive fine-scale structural variations, including abundant DUP, while INV are relatively less frequent (Lotter et al., 2022). Earlier studies suggest that SVs are subject to purifying selection, with INV experiencing stronger selective constraints than duplications due to their potentially greater deleterious impact on genome integrity and gene regulation (Zhou et al., 2019). Consistent with these findings, our analysis of *E. cloeziana* and *E. grandis* revealed a similar pattern, with an overall high level of collinearity and INV representing the smallest proportion of detected SVs (Fig. S2 and S3). These findings support the notion that, across *Eucalyptus* species, widespread fine-scale genomic SVs have occurred, contributing to genomic divergence and potentially driving adaptive differentiation.

4.4. Gene regulatory network underlying lignin, cellulose, and hemicellulose biosynthesis in E. cloeziana

Wood density is a key indicator of mechanical performance and is closely associated with wood properties, such as hardness, compressive strength parallel to the grain, and bending strength. A previous study has demonstrated that 17-year-old *E. cloeziana* presents a significantly greater basic density (0.67 g/cm^3) and vertical and parallel hardness (1133 and 1167 kgf, respectively) than *E. grandis* (0.59 g/cm^3, 531 kgf, and 664 kgf, respectively), which are 2.13-fold, 1.76-fold, and 1.20-fold greater in density, vertical hardness, and parallel hardness, respectively, and its bending strength (1032 kgf/cm^2) is 1.20-fold greater than that of *E. grandis* (858 kgf/cm^2) (Marini et al., 2022). Similarly, in 18-year-old trees, *E. cloeziana* presented a greater air-dry density (0.86 g/cm^3) and compressive strength (52.58 MPa) than *E. grandis* (0.67 g/cm^3 and 40.00 MPa, respectively) (Gonçalez et al., 2006). These results collectively highlight the mechanical superiority of *E. cloeziana* in terms of density, hardness, bending strength, and compressive strength over *E. grandis*, indicating its potential as a high-performance material for wood processing, construction, and furniture manufacturing. Wood mechanical properties are influenced by multiple factors, particularly the cellulose, hemicellulose, and lignin composition of the cell wall. Lignin, a polymer with high rigidity, enhances wood hardness (Nakano et al., 2015). The basic density of four-year-old *E. cloeziana* was 0.59 g/cm^3 , which was significantly higher than that of the *E. urophylla* × *E. grandis* clones EU1 (0.46 g/cm^3) and EU2 (0.45 g/cm^3). Additionally, the lignin content in *E. cloeziana* (32.28 %) surpassed that in EU1 and EU2 (both 30.31 %), suggesting that the genetic basis for lignin deposition contributes to its high hardness (Trugilho et al., 2015).

In our published study (Lei et al., 2025), we conducted a comparative analysis of wood properties between *E. cloeziana* and *E. urophylla* × *E. grandis*, using samples collected during the same period as in this study. The results demonstrated that *E. cloeziana* possesses significantly superior wood characteristics, with 27.7 % greater double wall thickness (10.92 vs 8.55 μm), 32.6 % higher basic density (0.57 vs 0.43 g/cm^3), 33.3 % greater absolute dry density (0.64 vs 0.48 g/cm^3), and 32.2 % elevated lignin content (174.01 vs 131.58 mg/g). These findings highlight *E. cloeziana*'s enhanced mechanical strength and commercial timber value. At the genomic level, *E. cloeziana* possesses 21 gene families involved in SCW biosynthesis, including 122 genes involved in lignin biosynthesis and 114 genes related to cellulose and hemicellulose biosynthesis—numbers that are greater than those in *E. grandis* (Fig. 6C). *PAL* encodes phenylalanine ammonia-lyase, a crucial enzyme in lignin biosynthesis, and comprises ten gene members in *E. cloeziana*. Similarly, gene families encoding *C4H*, *COMT*, *ferulate-5-hydroxylase* (*F5H*), and *HCT* are present in multiple copies, a phenomenon observed in other woody species, such as *S. suchowensis* (Wei et al., 2020), *P. trichocarpa* (Tuskan et al., 2006), *Mesua ferrea* (Sahu et al., 2023), and *Ferula assafoetida* (Amini et al., 2019), implying that the expansion of these gene families drives the lignin biosynthesis capacity. *4-coumarate 3-hydroxylase* (*C3H*), a cytochrome P450 family gene involved in lignin biosynthesis, is a single-copy gene in *E. cloeziana*, but there are multiple copies in *Salicaceae* species (Wei et al., 2020), suggesting possible species-specific regulation of lignin monomer composition.

Transcriptome analysis further confirmed that lignin biosynthesis genes are abundantly expressed in the xylem tissue of *E. cloeziana* (Fig. 6E), reinforcing their functional importance in wood formation. Cellulose is a major component of the plant cell wall, accounting for approximately 20 % of the primary wall and up to 50 % of the secondary wall (Baez et al., 2022). The *CESA* genes, which encode cellulose synthases essential for cellulose biosynthesis, are present in multiple copies in *A. thaliana* (Endler and Persson, 2011). Multiple copies of *CESA* genes are also found in *E. cloeziana*, exhibiting high expression levels in xylem tissue (Fig. 6D and F). Hemicellulose, which contributes to wall stability and integrity as another key component of the SCW, is predominantly present as xylan in broadleaf trees, and its biosynthesis is regulated by multiple gene families, including *UXS*, *IRX*, and *reduced wall acetylation* (*RWA*) (Julian and Zabotina, 2022). *UXS* encodes UDP-xylose synthase, and its mutation in *Arabidopsis* results in a collapsed xylem and reduced xylan content (Urbanowicz et al., 2014). *IRX* proteins function as accessory components of the xylan synthase complex, maintaining its structural integrity. Xylan acetylation is co-regulated by the *RWA*, *trichome birefringence-like* (*TBL*), and *altered xyloglucan 9* (*AXY9*) gene families (Lee et al., 2011), with *RWA* mediating the transmembrane transport of acetyl-CoA (Manabe et al., 2013). In *E. cloeziana*, we observed the expression of the *UXS*, *IRX*, and *RWA* genes in xylem tissue, with *IRX* and *RWA* showing greater expression in the xylem compared to the other tissues. Furthermore, the *UXS* gene family presented a greater copy number in *E. cloeziana* than in *E. grandis* (Fig. 6D and F).

Previous studies have demonstrated that manipulation of key genes in the lignin biosynthetic pathway can significantly alter lignin content and secondary cell wall structure in *Eucalyptus*. For example, downregulation of *Eucalyptus CCR1*, which encodes a pivotal enzyme catalyzing the conversion of cinnamoyl-CoA to cinnamaldehyde, led to transgenic hairy roots with markedly reduced lignin levels and thinner secondary cell walls, thereby validating its critical role in lignification and secondary cell wall formation (Plasencia et al., 2016). The most important enzyme of the phenylpropanoid pathway, 4-coumarate: coenzyme A ligase, is encoded by several homologous genes including *4CL1*, whose promoter functions as a tissue-specific regulatory element particularly active in the secondary xylem and is considered a promising candidate for driving foreign gene expression in plant xylem tissues (Hue et al., 2016). A previous study investigated the regulatory effects of *COMT* and *CCoAOMT* isolated from *E. urophylla*, where *Nicotiana tabacum* plants were separately transformed with sense *EuCOMT*, full-length interfering RNA of *EuCCoAOMT*, and both genes together. Compared with wild-type plants, transgenic lines showed less than 10 % changes in lignin and cellulose contents, while co-transformed plants exhibited a 57.38 % decrease in G-lignin content, indicating that suppressing *CCoAOMT* expression significantly inhibits G-lignin synthesis (Xiao et al., 2020). Additionally, suppression of *C3H* and *C4H* genes *E. urophylla* × *E. grandis* resulted in decreased total lignin content and dramatic shifts in monolignol composition, with *C3H* RNAi lines showing a significant increase in H-lignin content (18.3 %) and *C4H* RNAi lines maintaining levels (1.2 %) similar to control plants (Sykes et al., 2015). Our study revealed tissue-specific expression patterns of lignin, cellulose, and hemicellulose related genes, demonstrating regulation of SCW biosynthesis. In this study, we identified multiple structural variations affecting key lignin biosynthetic genes, including *C4H*, *4CL*, *CCR*, and *CCoAOMT*, which exhibit variations such as DUP, TRANS, and INV in *E. cloeziana* (Table S8). These SVs may influence gene dosage or gene regulation, potentially contributing to natural variation in lignin content. Given the established roles of *CCR*, *4CL*, *CCoAOMT* and *C4H* in lignin biosynthesis and xylem development in previous studies, such structural alterations could have functional consequences for wood formation and quality in *E. cloeziana*.

These findings suggest that the coordinated evolution and transcriptional regulation of SCW biosynthesis-related gene families underpin the superior hardwood properties of *E. cloeziana*. Building on these transcriptomic and genomic insights, and guided by findings from previous functional analyses, we plan to perform targeted gene function validation using approaches such as CRISPR-Cas9, RNAi, or gene overexpression in combination with hairy root transformation systems. Looking ahead, this integrated strategy holds significant promise for elucidating gene functions involved in wood formation and identifying molecular targets to improve wood quality traits for a range of industrial applications, including timber processing, construction, and high-value furniture manufacturing.

5. Conclusions

In this study, we generated and comprehensively annotated the high-quality nuclear and chloroplast genomes of *E. cloeziana*, which demonstrated high accuracy and completeness. Our study revealed more mRNAs and more types of ncRNAs than the publicly available genome annotation of *E. cloeziana*. We also provided functional annotations for *E. cloeziana* proteins, thus offering rich genetic resources for future studies. We systematically characterized the tissue-specific expression patterns of genes involved in lignin, cellulose, and hemicellulose biosynthesis pathways through transcriptomic analysis across multiple tissues. These findings offer important insights into the molecular mechanisms underlying wood formation in *E. cloeziana* and establish a foundation for future research on the transcriptional regulation of wood development and SCW biosynthesis. Our findings confirm that lignin biosynthesis-related genes, such as the well-characterized lignin-regulating genes *CCR*, *CCoAOMT*, *C4H*, and *4CL*, exhibit SVs (specifically DUP and TRANS) in *E. cloeziana*. Additionally, other candidate genes, including *HCT* and *CAD*, may contribute to lignin biosynthesis through their structural variations and warrant further functional validation to clarify their specific roles in the pathway. Moreover, this study provides genomic evidence for understanding the evolutionary relationships between *E. cloeziana* and other *Eucalyptus* species, laying the groundwork for future comparative genomics studies and molecular breeding efforts. Collectively, these results enhance our comprehension of the molecular mechanisms governing plant growth, development, and wood formation in *E. cloeziana*, providing support for the genetic improvement of this high-quality species.

CRediT authorship contribution statement

Meng Li: Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Wenfei Wu:** Writing – review & editing, Writing – original draft, Data curation. **Yi Mo:** Writing – review & editing, Writing – original draft. **Xian-Chen Geng:** Writing – review & editing, Writing – original draft. **Yuchong Fei:** Writing – review & editing, Writing – original draft. **Jiajing Xu:** Writing – review & editing, Writing – original draft. **Deyuan Lei:** Writing – review & editing, Writing – original draft. **Yanqi Cen:** Writing – review & editing, Writing – original draft. **Jun Ni:** Writing – review & editing, Writing – original draft. **Kuipeng Li:** Writing – review & editing, Writing – original draft. **Yunpeng Cao:** Writing – review & editing, Supervision, Methodology. **Zeng-Fu Xu:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Not applicable.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2025.121680.

Data availability

The raw sequence data reported in this paper have been deposited in the Genome Sequence Archive (GSA: CRA024838; https://ngdc.cncb.ac.cn/gsa/browse/CRA024838) in National Genomics Data Center, China National Center for Bioinformation (CNCB-NGDC). All of the genomes and genome annotation files, gene sequence query and download, gene expression and visualization, and commonly used bioinformatics analysis tools can be accessed from EucaMOD (http://eucalyptusggd.net/eucamod).

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